UNIVERSITY OF PISA

Lesson 2

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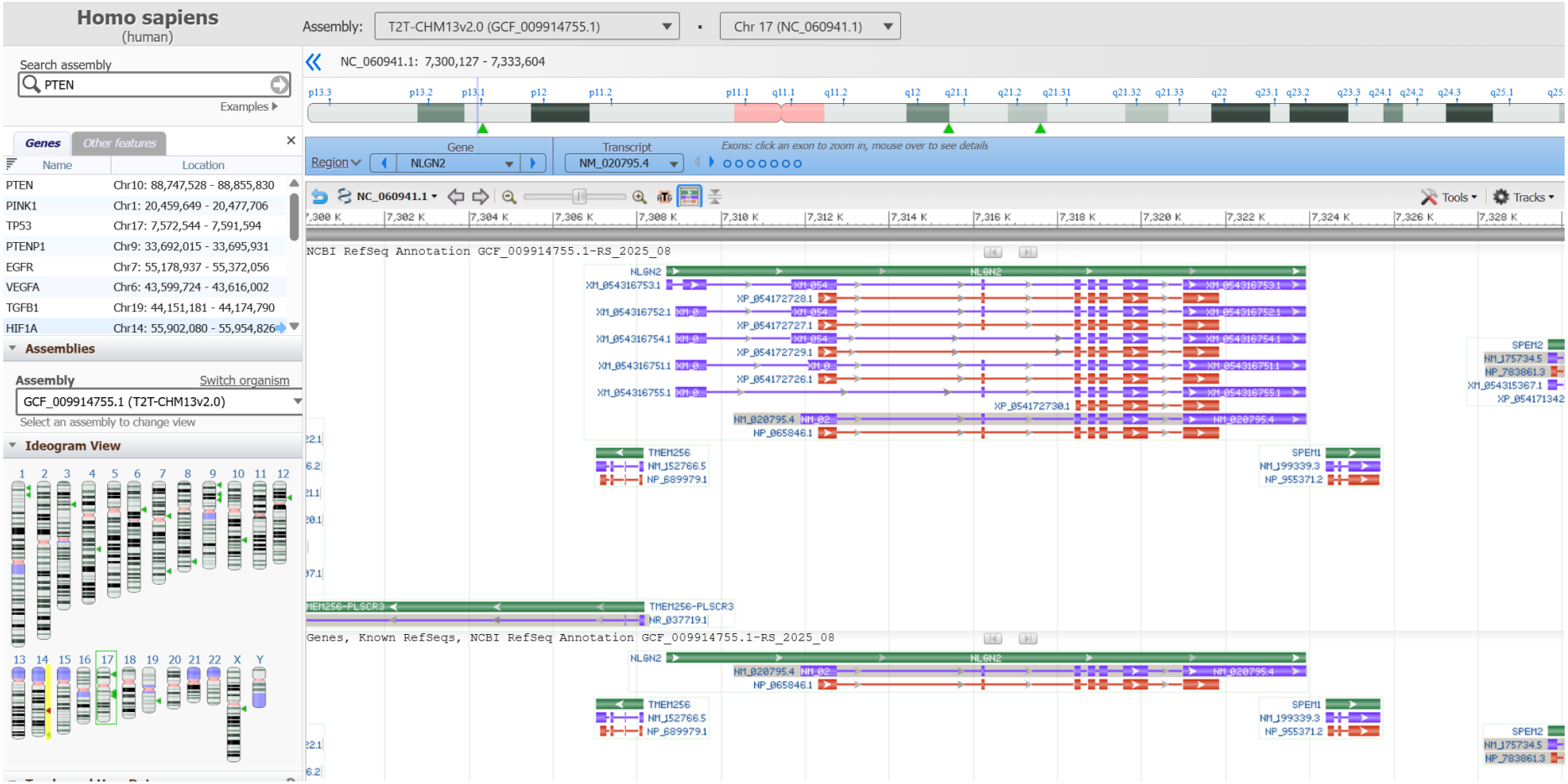
Where should I find sequences?

DATABASES

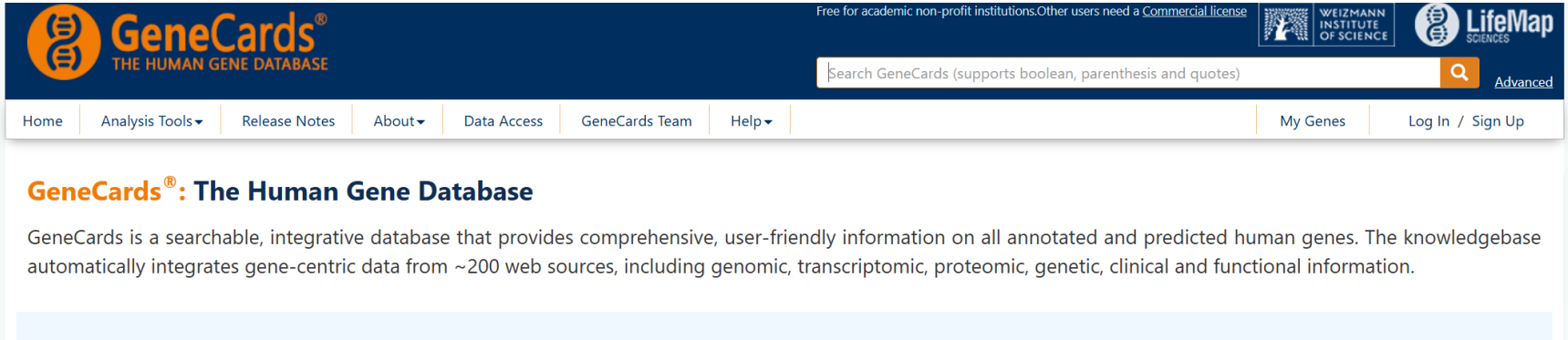
Where to find information? DNA/RNA

- Biological databases: among the objectives of Bioinformatics there is storing, indexing and searching biological information
  - Nucleaotide sequences databases:
    1. ENA www.ebi.ac.uk/ena
    2. GenBank https://www.ncbi.nlm.nih.gov/genbank/
    3. International Nucleotide Sequence Database Collaboration https://www.insdc.org/
  - Gene centric database
    - NCBI Gene https://www.ncbi.nlm.nih.gov/gene/
  - From GenBank deliver a comprehensive database from genome, transcriptome and proteome
    - NCBI RefSeq https://www.ncbi.nlm.nih.gov/refseq/about/
  - Human and mouse genome comprehensive annotation
    - Genecode https://www.gencodegenes.org/
  - Genome browsers
    - UCSC genome browser https://genome.ucsc.edu/
    - Ensembl https://www.ensembl.org/index.html

Genome Data Viewer



Gene cards


GeneCards®
THE HUMAN GENE DATABASE
Free for academic non-profit institutions.Other users need a Commercial license
WEIZMANN INSTITUTE OF SCIENCE
LifeMap SCIENCES
Search GeneCards (supports boolean, parenthesis and quotes)
Advanced
Home
Analysis Tools
Release Notes
About
Data Access
GeneCards Team
Help
My Genes
Log In / Sign Up
GeneCards®: The Human Gene Database
GeneCards is a searchable, integrative database that provides comprehensive, user-friendly information on all annotated and predicted human genes. The knowledgebase automatically integrates gene-centric data from ~200 web sources, including genomic, transcriptomic, proteomic, genetic, clinical and functional information.

Where to find information? Proteins

- Integrate proteins from different sources
  - NCBI proteins https://www.ncbi.nlm.nih.gov/home/proteins/
- Integrated and comprehensive repository on sequence and functional information. One of the most used!
  - UniProt www.uniprot.org
- Database that contains structural data information for proteins, nucleic acids and other complex assemblies
  - Protein Data Bank (PDB) www.rcsb.org

The FASTA format

- Simplified view of sequences
- Each record
  - starts with a “>” sign, followed by a description of the sequence
  - continues on multiple lines with the sequence itself
- Can include information on multiple sequences
  - DNA, RNA, Proteins

```
>NM_000546.6 Homo sapiens tumor protein p53 (TP53), transcript variant
CTCAAAAGTCTAGAGCCACCGTCCAGGGAGCAGGTAGCTGCTGGGCTCCGGGGACACTTTGCGTTCGGGC
TGGGAGCGTGCTTTCCACGACGGTGACACGCTTCCCTGGATTGGCAGCCAGACTGCCTTCCGGGTCACTG
CCATGGAGGAGCCGCAGTCAGATCCTAGCGTCGAGCCCCCTCTGAGTCAGGAAACATTTTCAGACCTATG
GAAACTACTTCCTGAAAACAACGTTCTGTCCCCCTTGCCGTCCCAAGCAATGGATGATTTGATGCTGTCC
CCGGACGATATTGAACAATGGTTCACTGAAGACCCAGGTCCAGATGAAGCTCCCAGAATGCCAGAGGCTG
CTCCCCCCGTGGCCCCTGCACCAGCAGCTCCTACACCGGCGGCCCCTGCACCAGCCCCCTCCTGGCCCCT
GTCATCTTCTGTCCCTTCCCAGAAAACCTACCAGGGCAGCTACGGTTTCCGTCTGGGCTTCTTGCATTCT
GGGACAGCCAAGTCTGTGACTTGCACGTACTCCCCTGCCCTCAACAAGATGTTTTGCCAACTGGCCAAGA
CCTGCCCTGTGCAGCTGTGGGTTGATTCCACACCCCCGCCCGGCACCCGCGTCCGCGCCATGGCCATCTA
CAAGCAGTCACAGCACATGACGGAGGTTGTGAGGCGCTGCCCCCACCATGAGCGCTGCTCAGATAGCGAT
GGTCTGGCCCCTCCTCAGCATCTTATCCGAGTGGAAGGAAATTTGCGTGTGGAGTATTTGGATGACAGAA
ACACTTTTCGACATAGTGTGGTGGTGCCCTATGAGCCGCCTGAGGTTGGCTCTGACTGTACCACCATCCA
CTACAACTACATGTGTAACAGTTCCTGCATGGGCGGCATGAACCGGAGGCCCATCCTCACCATCATCACA
CTGGAAGACTCCAGTGGTAATCTACTGGGACGGAACAGCTTTGAGGTGCGTGTTTGTGCCTGTCCTGGGA
GAGACCGGCGCACAGAGGAAGAGAATCTCCGCAAGAAAGGGGAGCCTCACCACGAGCTGCCCCCAGGGAG
CACTAAGCGAGCACTGCCCAACAACACCAGCTCCTCTCCCCAGCCAAAGAAGAAACCACTGGATGGAGAA
```

Filename extension [edit]

There is no standard filename extension for a text file containing FASTA formatted sequences. The table below shows each extension and its respective meaning.

Extension	Meaning	Notes
fasta, fas, fa[13]	generic FASTA	Any generic FASTA file
fna	FASTA nucleic acid	Used generically to specify nucleic acids
ffn	FASTA nucleotide of gene regions	Contains coding regions for a genome
faa	FASTA amino acid	Contains amino acid sequences
mpfa	FASTA amino acids	Contains multiple protein sequences
frn	FASTA non-coding RNA	Contains non-coding RNA regions for a genome, e.g. tRNA, rRNA

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Transcript annotation files

- The GFF (General Feature Format), GTF (General Transfer Format) is identical to GFF2, now deprecated (but still used), GFF3.
- GFF2 can only represent 2 level feature hierarchies, while GFF3 can support arbitrary levels.
- GFF files are plain text, 9 column, tab-delimited files

```
##gff-version 3
ctg123 . mRNA            1300  9000  .  +  .  ID=mrna0001;Name=sonichedgehog
ctg123 . exon            1300  1500  .  +  .  ID=exon00001;Parent=mrna0001
ctg123 . exon            1050  1500  .  +  .  ID=exon00002;Parent=mrna0001
ctg123 . exon            3000  3902  .  +  .  ID=exon00003;Parent=mrna0001
ctg123 . exon            5000  5500  .  +  .  ID=exon00004;Parent=mrna0001
ctg123 . exon            7000  9000  .  +  .  ID=exon00005;Parent=mrna0001
```

The GenBank format

- General information: IDs, description, publications
- Features
- Sequence

```
LOCUS       NM_000546                2512 bp    mRNA    linear   PRI 18-AUG-2022
DEFINITION  Homo sapiens tumor protein p53 (TP53), transcript variant 1, mRNA.
ACCESSION   NM_000546
VERSION     NM_000546.6
KEYWORDS    RefSeq; MANE Select.
SOURCE      Homo sapiens (human)
  ORGANISM  Homo sapiens
            Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
            Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
            Catarrhini; Hominidae; Homo.
REFERENCE   1  (bases 1 to 2512)
  AUTHORS   Mirgayazova R, Khadiullina R, Chasov V, Mingaleeva R, Miftakhova R,
            Rizvanov A and Bulatov E.
  TITLE     Therapeutic Editing of the TP53 Gene: Is CRISPR/Cas9 an Option?
  JOURNAL   Genes (Basel) 11 (6), E704 (2020)
   PUBMED   32630614
  REMARK    GeneRIF: Therapeutic Editing of the TP53 Gene: Is CRISPR/Cas9 an
            Option?
            Review article
            Publication Status: Online-Only
```

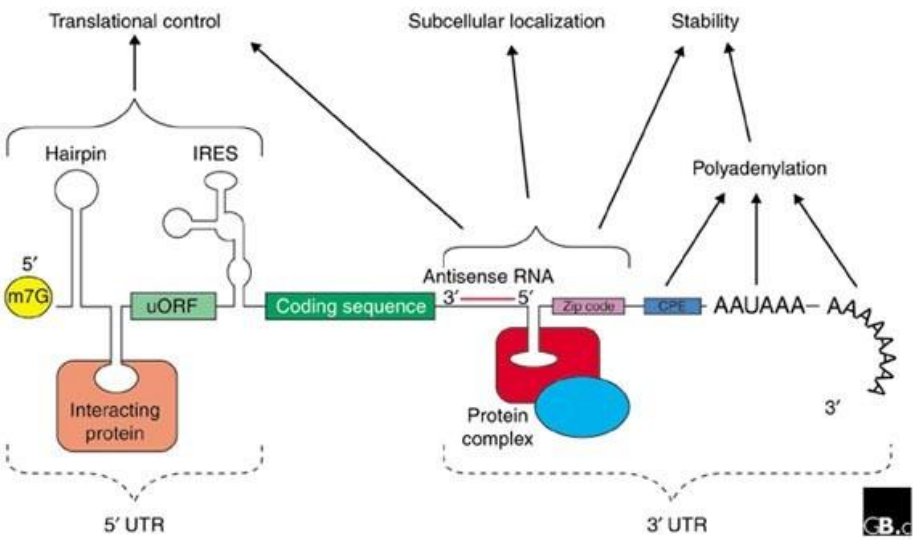
```
ORIGIN
        1 ctcaaaagtc tagagccacc gtccagggag caggtagctg ctgggctccg gggacacttt
       61 gcgttcgggc tgggagcgtg ctttccacga cggtgacacg cttccctgga ttggcagcca
      121 gactgccttc cgggtcactg ccatggagga gccgcagtca gatcctagcg tcgagccccc
      181 tctgagtcag gaaacatttt cagacctatg gaaactactt cctgaaaaca acgttctgtc
      241 ccccttgccg tcccaagcaa tggatgattt gatgctgtcc ccggacgata ttgaacaatg
      301 gttcactgaa gacccaggtc cagatgaagc tcccagaatg ccagaggctg ctccccccgt
      361 ggcccctgca ccagcagctc ctacaccggc ggcccctgca ccagccccct cctggcccct
      421 gtcatcttct gtcccttccc agaaaaccta ccagggcagc tacggtttcc gtctgggctt
      481 cttgcattct gggacagcca agtctgtgac ttgcacgtac tcccctgccc tcaacaagat
```

```
FEATURES             Location/Qualifiers
     source          1..2512
                     /organism="Homo sapiens"
                     /mol_type="mRNA"
                     /db_xref="taxon:9606"
                     /chromosome="17"
                     /map="17p13.1"
     gene            1..2512
                     /gene="TP53"
                     /gene_synonym="BCC7; BMFS5; LFS1; P53; TRP53"
                     /note="tumor protein p53"
                     /db_xref="GeneID:7157"
                     /db_xref="HGNC:HGNC:11998"
                     /db_xref="MIM:191170"
     exon            1..114
                     /gene="TP53"
                     /gene_synonym="BCC7; BMFS5; LFS1; P53; TRP53"
                     /inference="alignment:Splign:2.1.0"
```

UTR

- Un-Translated Regions
- 3’ UTR: a segment of mRNA, after the coding region. It is critical for regulating gene expression by influencing mRNA stability, localization within the cell, and translation of the mRNA into protein. The 3’ UTR achieves these functions by binding to regulatory molecules like microRNAs and RNA-binding proteins.

- 5’ UTR: segment of mRNA located before the start codon (AUG), which plays a crucial role in regulating translation by controlling ribosome binding, mRNA stability, and protein synthesis efficiency through various structural and sequence elements.



Mignone, F., Gissi, C., Liuni, S. *et al.* Untranslated regions of mRNAs. *Genome Biol* **3**, reviews0004.1 (2002). https://doi.org/10.1186/gb-2002-3-3-reviews0004

Questions?

Homework

- Consider the TP53 gene in human. From NCBI Gene get the genomic sequence. Retrieve the list of protein sequences derived from each isoform.
- For the same gene, get from NCBI RefSeq the coding sequence and the protein sequences for human, chimp and mouse.
- In Ensembl Genome Browser visualize the information related to the human TP53 gene
  - The longest transcript and get the coding sequence;
  - How many splice variants do you find for this gene? How many protein coding?
  - Get the evolutionary tree of TP53
  - Find a germline (identified by *rs* prefix) and somatic (identified with *COSM* prefix) SNP occurring within the coding sequence of the gene.
  - Get the 3'UTR and the 5'UTR sequences for this gene

Alignment

COMPARISON OF SEQUENCES

Sequence analysis - pattern matching

- Why is important?
  - Analysis of biological sequences, e.g. DNA or proteins
    - Similar patterns can imply similar function: e.g. proteins domains or DNA sequence that binds regulatory proteins
    - It can help to reconstruct genome/gene/protein evolution
  - All search problems are solved with pattern matching algorithms
    - Exact pattern matching:
      - Given a sequence of length ***s*** and a pattern of length ***p*** ($p <= s$), find the occurrences of that pattern exactly as it is
    - Approximate pattern matching:
      - Given a sequence of length ***s*** and a pattern of length ***p***, find the occurrences of that pattern but allowing for some errors - mismatches

Exact pattern matching

- Naive algorithm
  - Simply parse the sequence and for each character parse the pattern
- Time complexity
  - Worst case: for each character in the sequence we need to parse all characters in the pattern
    - f(s*p) operations - O(s*p)

Knuth-Morris-Pratt (KMP) algorithm

- Preprocess step: analyze the pattern to construct the **LPS** array -- “Longest Proper Prefix that is also Suffix”
- Use this array to skip some comparisons

```
TTATCATCAGATCATCGGATGTT        P: ATCATCGGATG
  ATCATCGGATG

TTATCATCAGATCATCGGATGTT
   ATCATCGGATG

TTATCATCAGATCATCGGATGTT
      ATCATCGGATG
```

Naive

KMP

Let’s code

Classes of patterns - regular expressions

- Syntax of a regular expression
  - alphabet: characters that are matched as they are in the regular expression
    - for DNA we will have **A, C, G, T**
    - for RNA we have **A, C, G, U**
    - for proteins we have the 20 amino acids
  - metacharacters:
    - allow for selecting, filtering and repeating characters in the alphabet to create a pattern
    - the exact set depends on the matching engine used - some standards exist
      - Python `re` module: **. ^ $ * + ? { } [ ] \ | ( )**
      - https://docs.python.org/3/howto/regex.html#regex-howto

IUPAC symbols for nucleotides and aminoacids

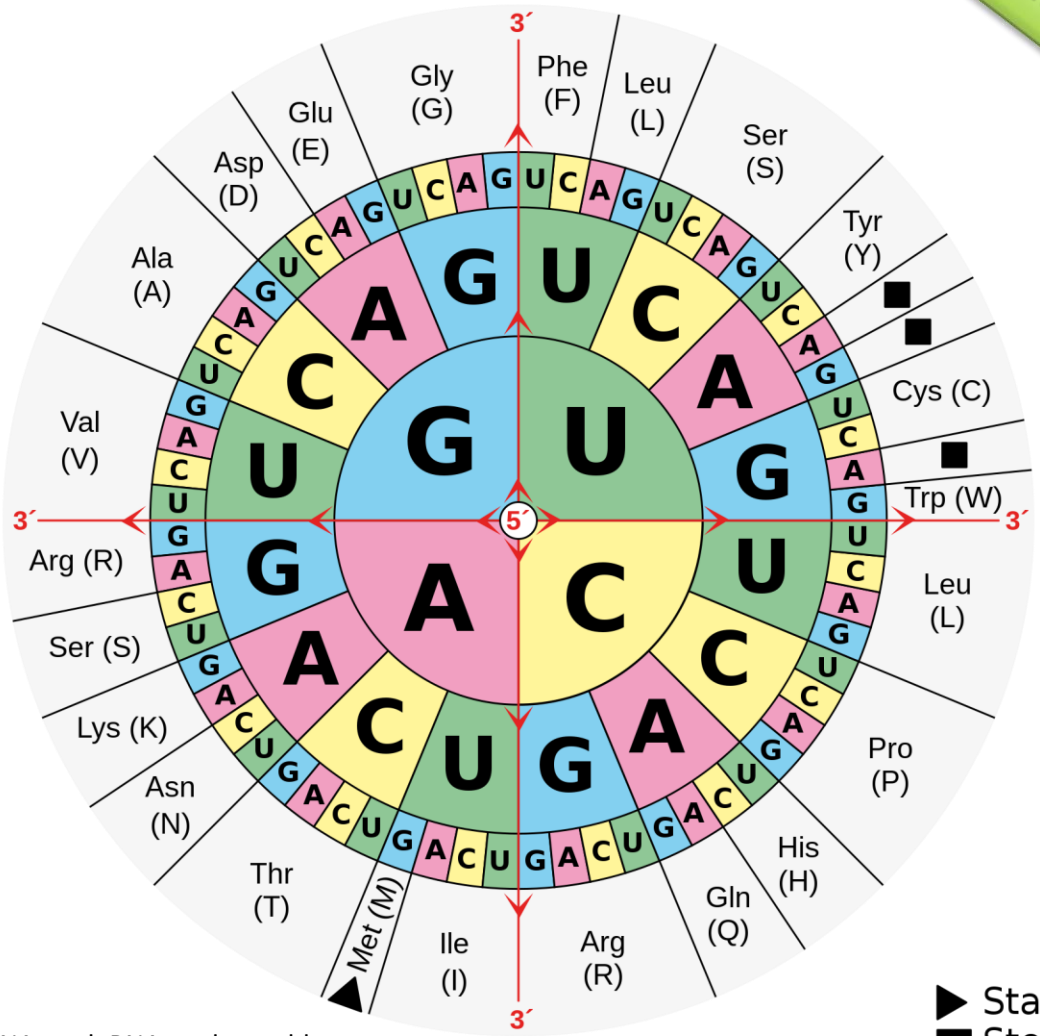
IUPAC nucleotide code	Name	Nucleotide Represented
A	Adenine	A
C	Cytosine	C
G	Guanine	G
T	Thymine	T
U	Uracil	U
R	Purine	A or G
Y	Pyrimidine	C or T
S	Strong	G or C
W	Weak	A or T
K	Keto	G or T
M	Amino	A or C
B	Not A	C or G or T
D	Not C	A or G or T
H	Not G	A or C or T
V	Not T	A or C or G
N	any base	A, T, C, G

IUPAC code	Three letter code	Amino acid
A	Ala	Alanine
C	Cys	Cysteine
D	Asp	Aspartic Acid
E	Glu	Glutamic Acid
F	Phe	Phenylalanine
G	Gly	Glycine
H	His	Histidine
I	Ile	Isoleucine
K	Lys	Lysine
L	Leu	Leucine
M	Met	Methionine
N	Asn	Asparagine
P	Pro	Proline
Q	Gln	Glutamine
R	Arg	Arginine
S	Ser	Serine
T	Thr	Threonine
V	Val	Valine
W	Trp	Tryptophan
Y	Tyr	Tyrosine

Codons to aa

Let's code

A triplet refers to the genetic code where a sequence of three nucleotides (called a codon) specifies a particular amino acid



Nearly universal… not completely

▶ Start
■ Stop

https://en.wikipedia.org/wiki/DNA_and_RNA_codon_tables

Classes of patterns - regular expressions

RE most used meta-charaters

- . (dot) matches any character (e.g. … will match any string of length 3)

- * means zero or more repetitions of a pattern
- + one or more repetitions of a pattern
- ? Zero or one repetition
- {n} exactly *n* repetitions
- {m,n} between *m* and *n* repetitions, where *m* and *n* are integers and *n>= m*

modifiers

- **[A-Z]** matches any upper-case letter;
- **[a-z]** matches any lower-case letter;
- **[0-9]** matches any digit;
- **[ACTGactg]** matches a DNA sequence
- If **^** is put before a list, it means negation. **[^0-9]** means "not digits"
- If **^** is put before a character, it means that the line need to start with that character
- **** - define special sets of characters:
  - **\d** - digits, **\s** - white spaces, **\w** - any alphanumeric character
- **|** - OR operator - combine two or more REs - the first match is considered
- **()** - allows to delimit parts of the RE on which to apply metacharacters or operators
  - **ab*** different from **(ab)***

They can be applied to any character or group of charater

Homework

- Write a function that translate a DNA sequence into an amino acid sequence.
- Create a function that translate codons into a single letter aminoacid using the RE
- Most introns can be recognized by their consensus sequence which is defined as: GT…TACTAAC..AC, where … mean an unknown number of nucleotides (between 1 and 10).
  - Write a Python function that, given a DNA sequence, checks if it contains an intron, according to this definition. The result should be a list with all initial positions (empty list if there are none)
  - Download from NCBI or Ensambl genome browser a gene sequence and test the function